

Wideband Rx

Lineup and System Specs

Previous points of discussion

- DC cancellation (Zero DC for now)
- Filter Specs:
 - Rejection at 300MHz = 25dB
 - OIP2 = 55dB (for filter and VGA)
- Blocker at Psens + 6dB
- Effect of increasing Enob of ADC on SNDR

Previous points of discussion

- RF amplifier is added between DSAs
- Output Blocker power is calculated at the input of ADC

Proposed Line-up

System Description Component Name	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
	LNA	Attenuator	Amplifier	Attenuator	Mixer
Gain (dB)	22	-3	8	-2	-7
NF (dB)	1	3	2	2	7
P1dB (dBm)	15	15	15	15	6
IP3 (dBm)	25	25	25	25	16

System Description Component Name	Stage 6	Stage 7
	Filter	V G A
Gain (dB)	0	15
NF (dB)	12	3
P1dB (dBm)	6	10
IP3 (dBm)	16	20

OUTPUTS

Parameter	Output
Gain (dB)	33.00
NF (dB)	1.87
P1dB (dBm)	8.77
IP3 (dBm)	18.77

Cascade analysis (Max gain)

Proposed Line-up

System Description Component Name	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
	LNA	Attenuator	Amplifier	Attenuator	Mixer
Gain (dB)	22	-18	8	-5	-7
NF (dB)	1	18	2	5	7
P1dB (dBm)	15	15	15	15	6
IP3 (dBm)	25	25	25	25	16

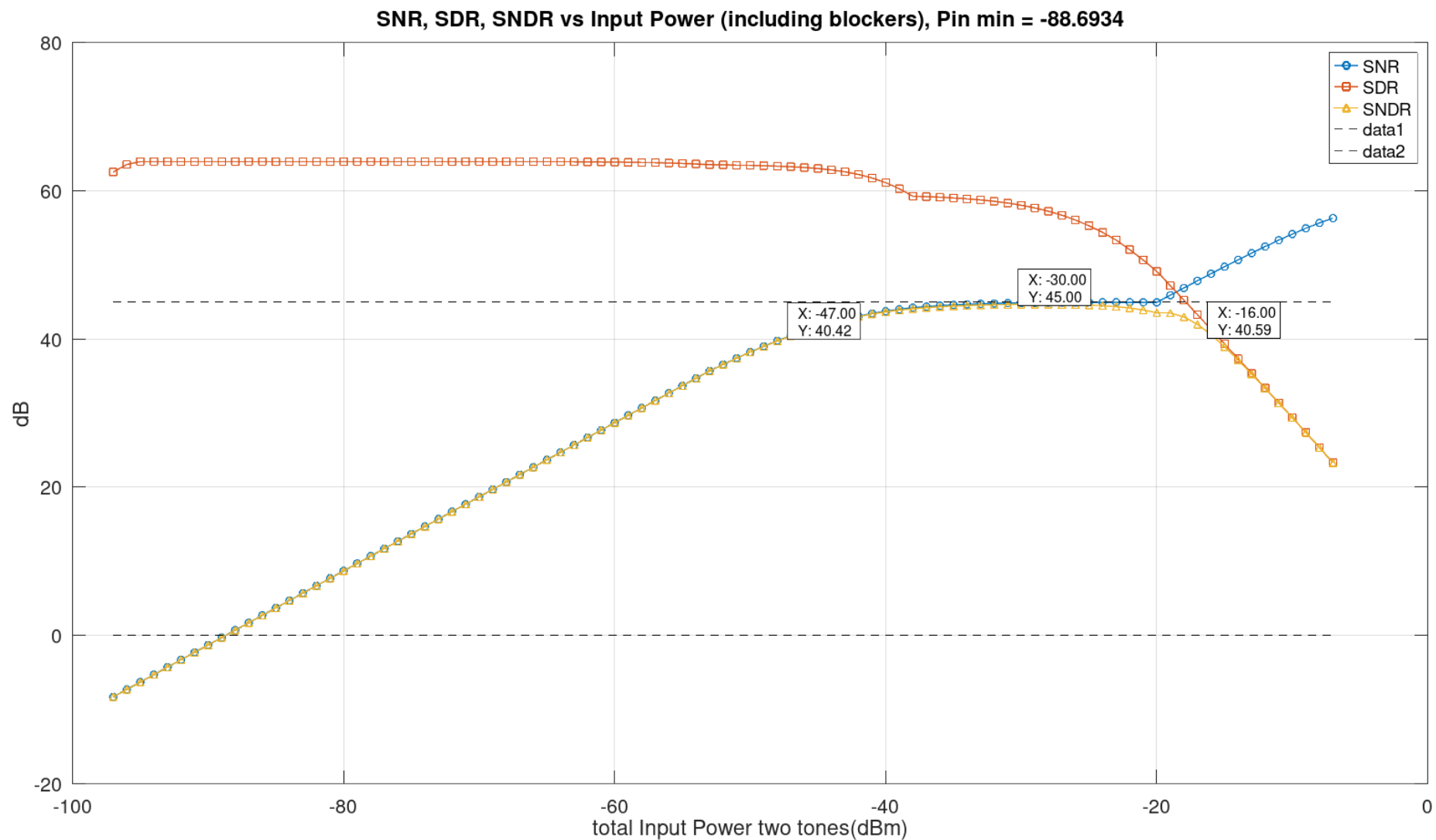
System Description Component Name	Stage 6	Stage 7
	Filter	V G A
Gain (dB)	0	0
NF (dB)	12	18
P1dB (dBm)	6	10
IP3 (dBm)	16	20

OUTPUTS

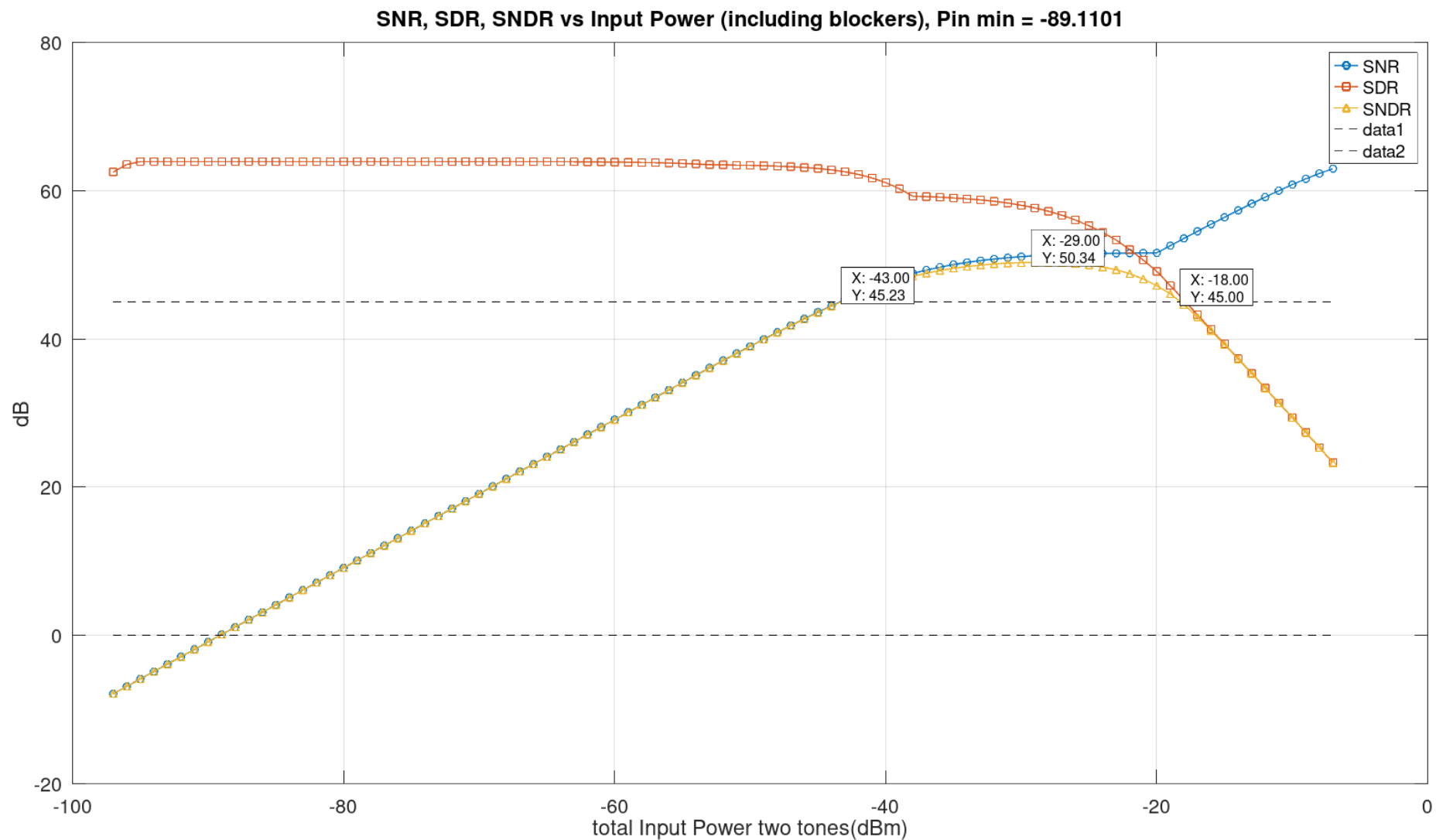
Parameter	Output
Gain (dB)	0.00
NF (dB)	19.02
P1dB (dBm)	-8.03
IP3 (dBm)	1.97

Cascade analysis (Min. gain)

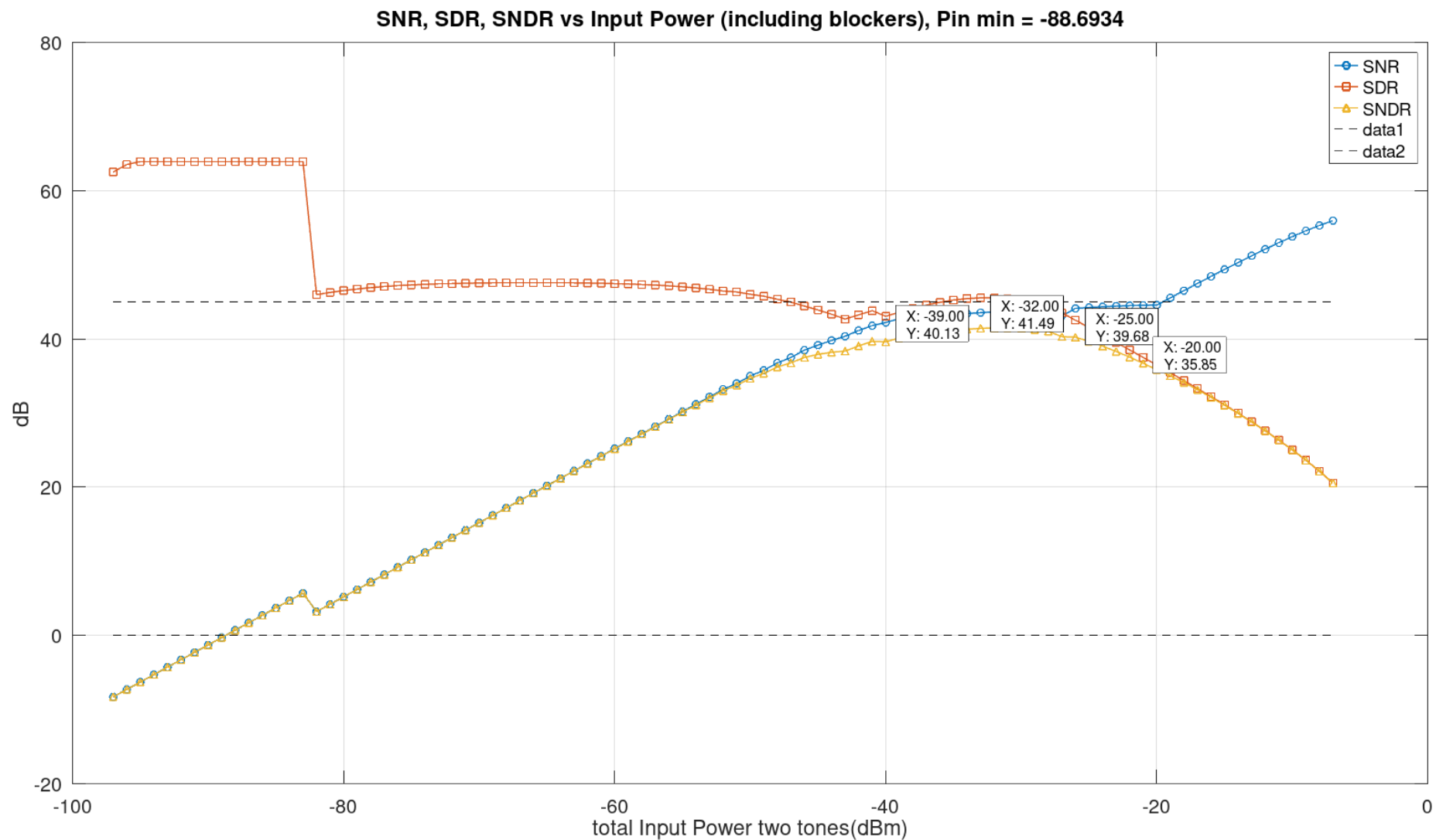
SNR, SDR and SNDR without blockers



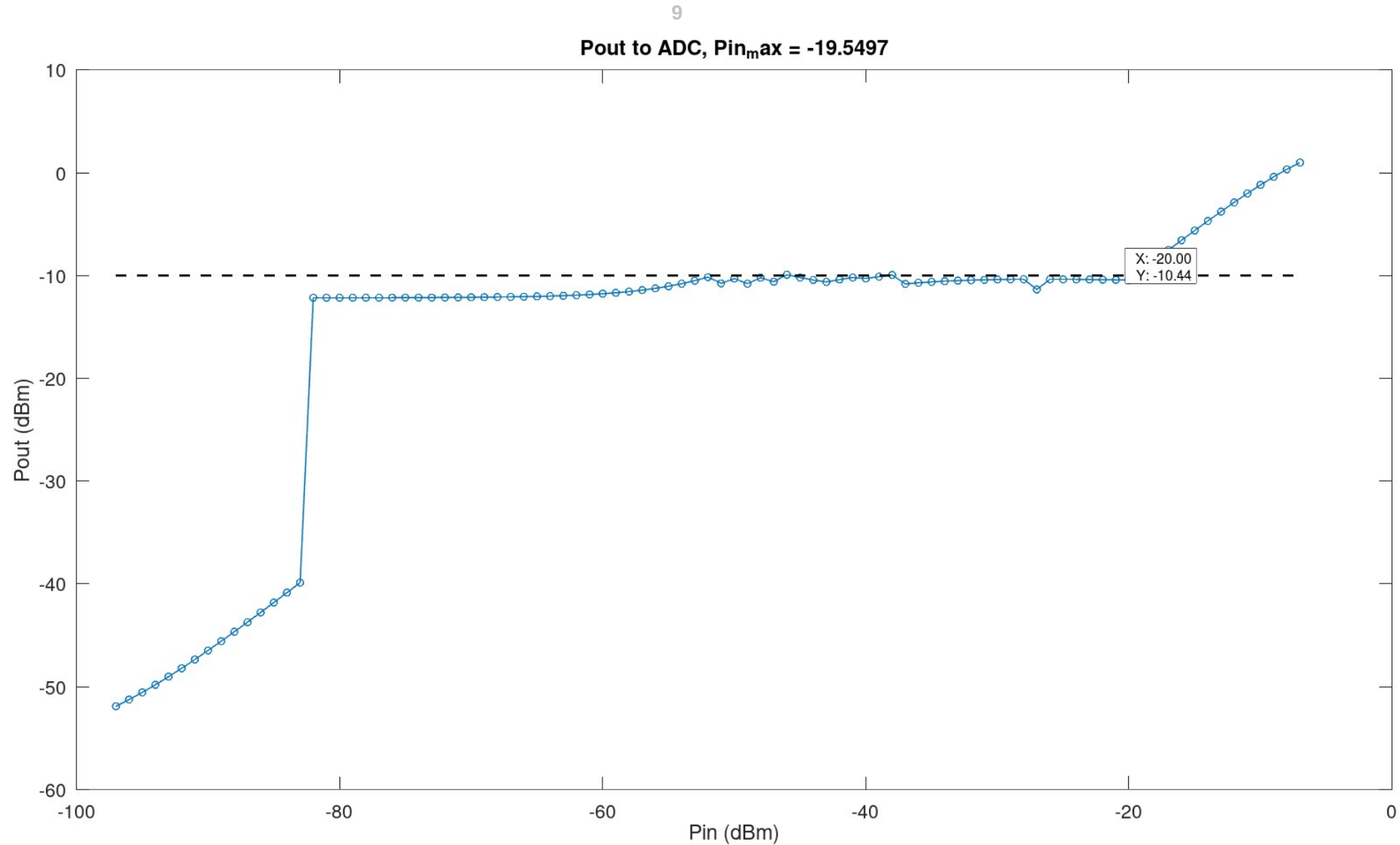
SNR, SDR and SNDR without blockers (ADC enob = 12bits)



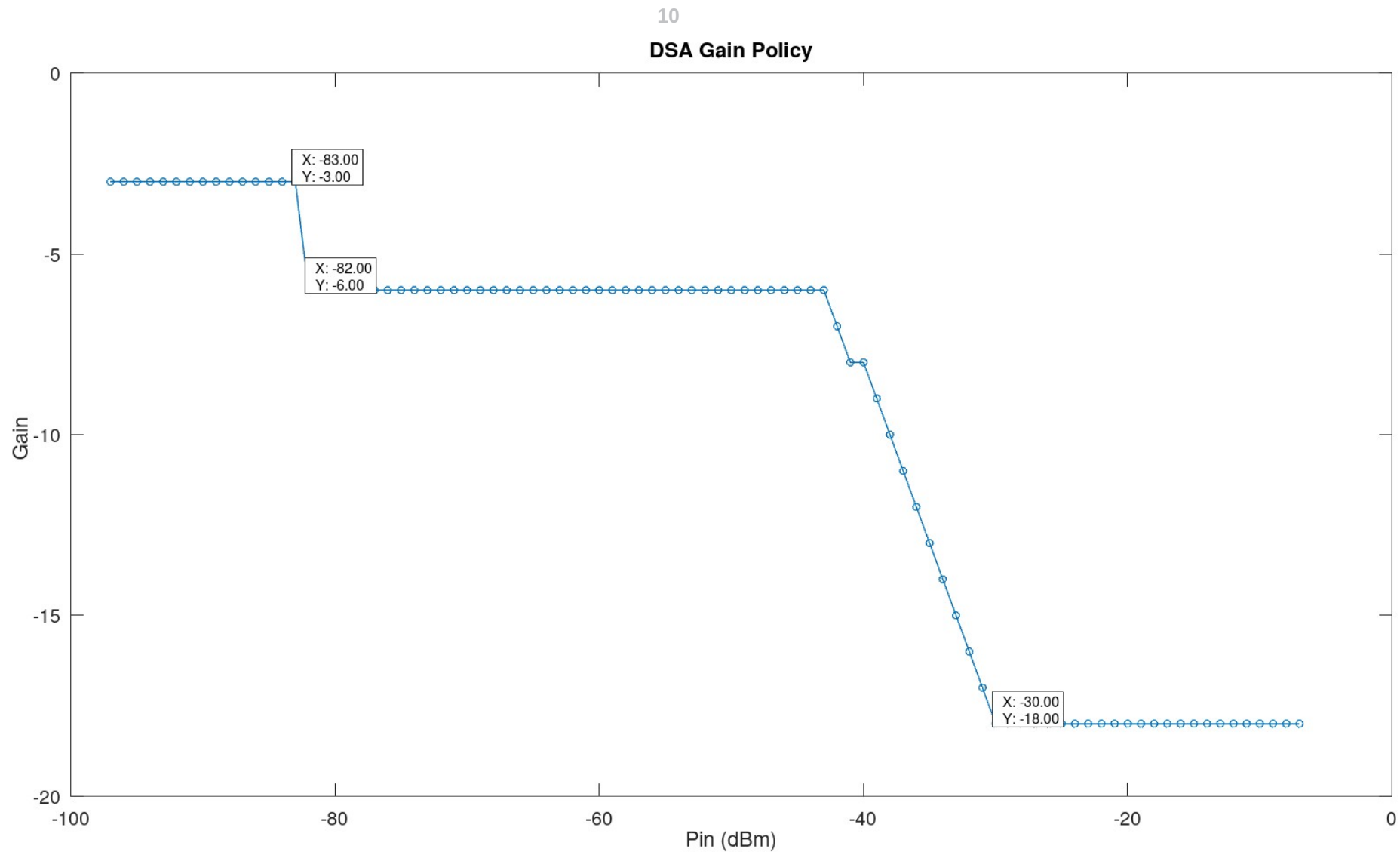
SNR, SDR and SNDR without blockers (With blocker and IP2)



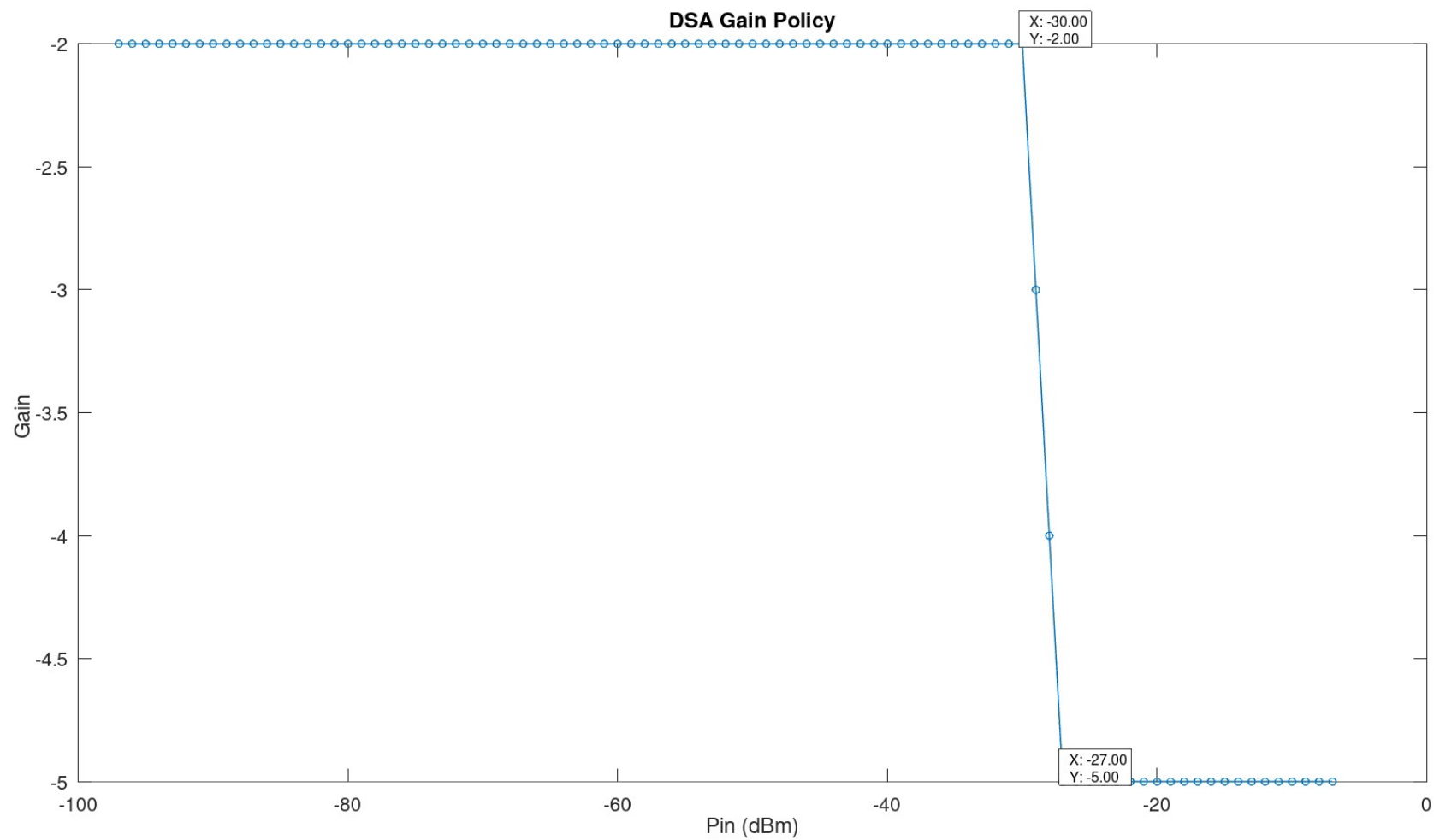
Maximum Input power for ADC



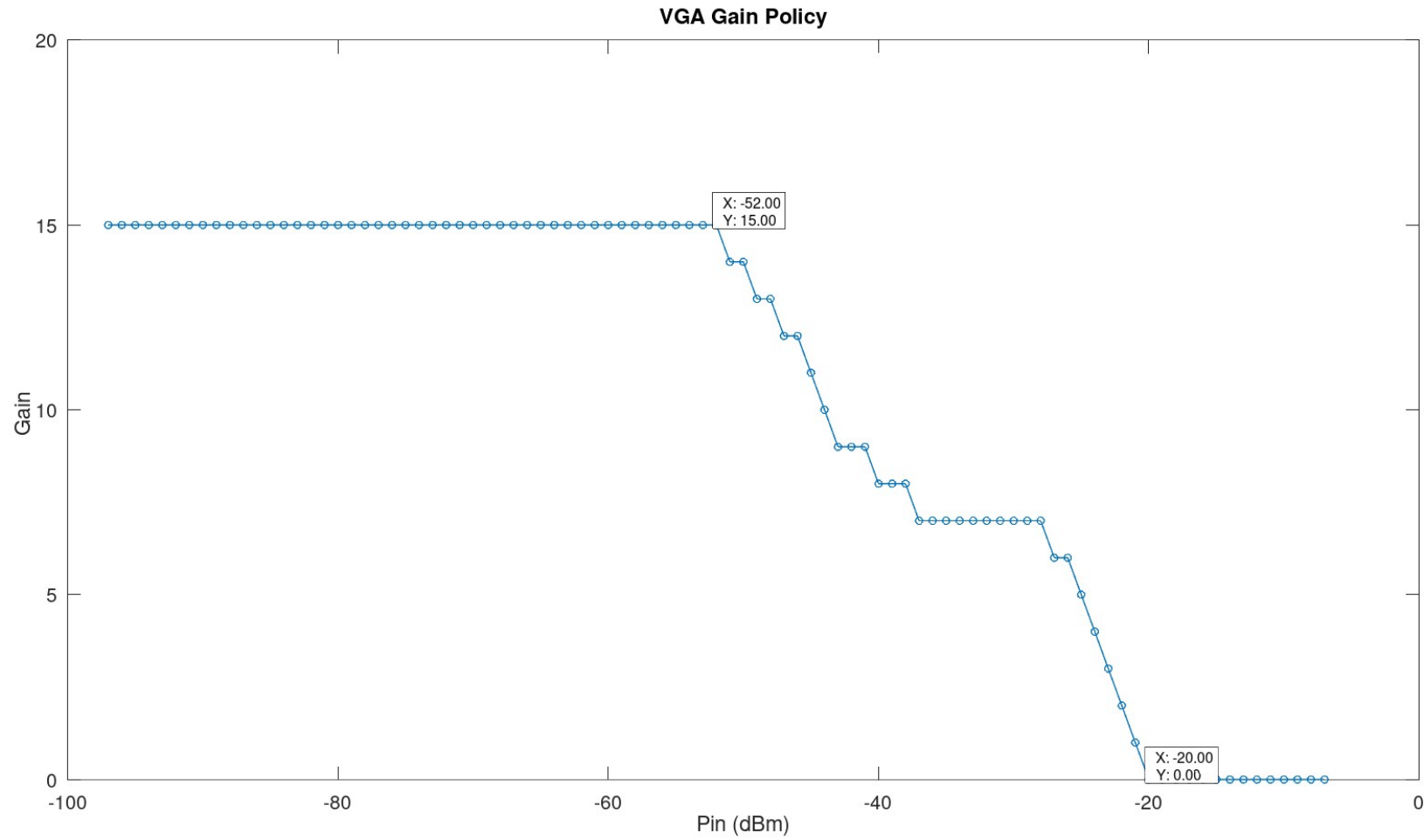
Gain Policy Under Blocking (First DSA)



Gain Policy Under Blocking (Second DSA)

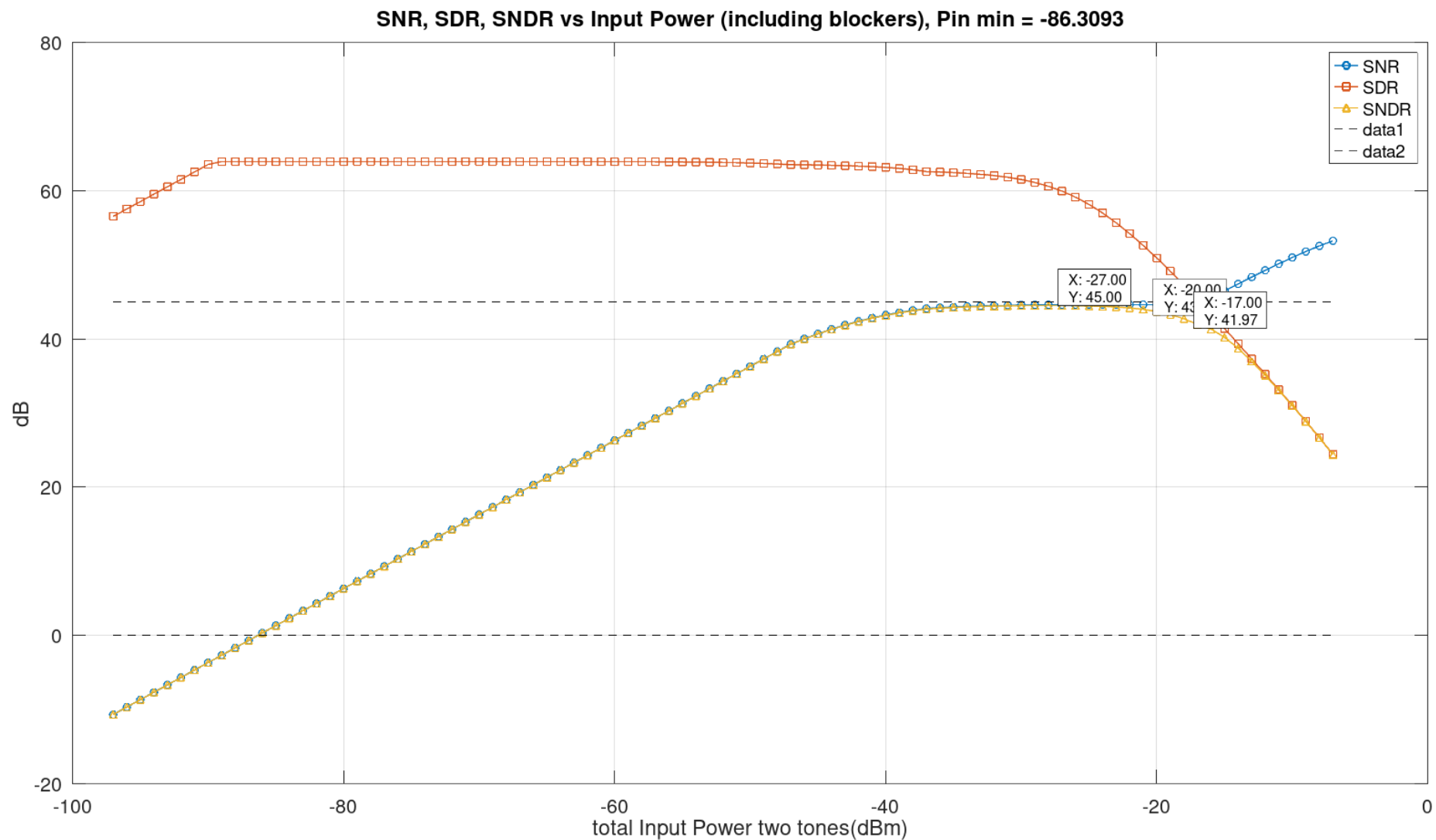


Gain Policy Under Blocking (VGA)



Comparison Between With and Without RF Amp

13



Comparison Between With and Without RF Amp

